Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Automation of rack hanging and unhanging of parts for coating applications using robots			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and materials		Machinery and materials	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics and Automation Machinery	Robot Vision and Production Automation Technologies	Robotics and Automation Machinery	Robot Control and Intelligence Technologies

Concept of Proposed Technology	**Core concept**
	
	An integrated robotic system that automates the part handling for rack-based coating processes: from extracting parts from bins, hanging them onto a rack, taking them from the rack to re-storing them in a bin. Emphasis is on robust gripping, mounting and placing. The system should be able to handle moving and swinging racks.
	
	Main functions
	
	• Handling workflow
	  ○ Pick parts from bins (ordered or chaotic storage)
	  ○ Mount parts onto a rack (hanging or clamping) using visual information			

- Grasp parts from the rack after coating
- Place parts into a load carrier (e.g., a bin)
- Perform the task robustly despite moving/swinging racks or conveyors
- Collect process data for monitoring

- Automatic picking of parts from bins or load carriers, both in ordered and in chaotic storage situations
  - Fraunhofer IPA does already have extensive experience in this area and industrial proven solutions (1)
- Mounting parts onto racks (hanging or clamping) based on visual information and geometric models
  - Fraunhofer IPA has already tested this application in a simple lab setup (2) and has done even more research in simulation
- Removing coated parts from racks by detecting the individual parts
- Placing the coated part into a suitable container
  - Fraunhofer IPA does already have this technology (3)
- Adapting to different parts, hooks, and rack layouts without extensive reprogramming
- Perceiving and tracking the position and motion of racks and conveyors to ensure robust handling even under movement
- Perception: Robust identification, localization, and tracking of parts, hooks, and racks in cluttered industrial environments
- Learning-based grasping and placement: Grasping, picking, and placing across different parts, including randomly deformed hooks
- Motion planning and control
  - High-precision mounting under uncertainty in perception and models
  - Executing collision-free, precise motions in confined, cluttered industrial environments
- Handling dynamic scenes: Robust perception and manipulation with moving and swinging targets
- Integration and validation
  - System integration, safety compliance, staged testing, and performance metrics (uptime, throughput)
  - End-to-end validation of a prototype in realistic painting or galvanizing lines

Potential uses

- Primary application: automatic rack handling in coating processes (e.g., painting, galvanizing) with integrated process monitoring
- Other applications: rack-based handling in other manufacturing or treatment lines (surface finishing, heat treatment, assembly lines)
- Benefits: reduced manual work and exposure to coatings, improved repeatability and

(max 10 pages)

	traceability, enhanced safety and quality management
Rationale for Joint R&D Project	Application partners are needed to provide real use cases from coating lines, supply relevant test materials (bins, racks, parts, carriers), and actively participate in evaluating and testing the automated rack-handling solution. Their involvement enables realistic validation, feedback on performance (throughput, uptime, safety, and quality), and rapid tailoring of perception, grasping, and control strategies to actual production conditions. The partner is also needed for the integration and validation of a on-site field-test. Ideally, the partner could also support in hardware development, like versatile grippers.
Key Words	1. Automated rack hanging 2. Cognitive robotics 3. 3D perception 4. Learning-based grasping and placement 5. Bin Picking and Bin Packing

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6-7)
R&D Objectives	• Perception of hooks, racks and parts using 3D sensor data - Robust identification, localization, and tracking in cluttered industrial environments. - Approaches: 3D vision (RGB-D, LiDAR, or stereo), data fusion, pose estimation, and tracking. • Learning-based grasping and placement strategies on the rack - Generalizable policies for grasping, picking, and placing across parts, including randomly deformed hooks. - Approaches: Reinforcement Learning with sim-to-real transfer; offline data collection and safe exploration. - Considerations: tactile feedback, multiple gripper configurations, and safety. • Handling dynamic scenes with moving and swinging targets - Robust perception and manipulation under motion. - Approaches: Real-time tracking, motion forecasting, visual servoing, and compliant control. • Motion planning and control for precise clamping and hanging under uncertainty - High-precision mounting despite perception and model uncertainties. - Approaches: Trajectory optimization and MPC, visual servoing, uncertainty handling. • Integration and validation in realistic industrial scenarios (painting / galvanizing lines) - Hardware design, system integration, safety compliance, staged testing, and performance metrics (uptime, throughput) • End-to-end validation of a prototype.			
R&D Target Specifications	• Objective: Develop an integrated robot system for automated rack handling in coating lines (bin extraction to re-storage) with end-to-end process monitoring and robust perception/manipulation for moving/swinging racks. • Cycle time for each process (racking and unracking) ≤ 8 s (max 16 s) • Success rate: - racking success ≥ 95%; unracking success ≥ 95%; in simulation - racking success ≥ 80%; unracking success ≥ 90%; in real setup, depending on the state of the racks, e.g. deformed hooks - picking success (from bin) ≥ 95% • Validation and interfaces: end-to-end prototype validated on realistic coating lines with data logging			

(max 10 pages)

R&D Plan

Phase I: Pre-study (months 0 – 6)

- Objective: Technical pre-study to confirm feasibility; prepare for Phase II
- Key tasks:
  - use-case refinement, rack/part cataloging
  - Small lab proof-of-concept on fixed racks.
  - Early sim-to-real planning and testbed setup (fixed-rack experiments).
- Deliverables:
  - Requirements document and success criteria.
  - Lab proof-of-concept results (fixed-rack perception and manipulation).
- Milestones:
  - Lab proof-of-concept for perception and grasping on fixed racks.

Phase II: Main R&D (months 7 – 36)

- WP 2.1: Perception, grasping, fixed racks; data collection; lab-grade prototypes
  - Months 7–12
  - Key Tasks:
    - Improve robust 3D perception pipeline; pose estimation for hooks and parts on racks
    - Collect diverse datasets; baseline offline learning
    - Basic grasping and mounting on fixed racks
  - Milestones:
    - Perception module validated on fixed rack tests
    - Dataset matured and labeled
  - Deliverables:
    - Lab perception and manipulation prototypes
    - Initial dataset and metrics report
- WP 2.2: Small end-to-end line mock-up in lab
  - Months 13–18
  - Key Tasks:
    - Hardware design, especially versatile gripper(s)
    - End-to-end workflow on line mock up (bin → rack → coating → rack → bin)
  - Milestones:
    - End-to-end demo on line mock up
  - Deliverables:

	▪ Integrated line mock up demo; data-logging framework; process-monitoring dashboard ○ WP 2.3: Learning-based policies ▪ Months 18–25 ▪ Key Tasks: ▪ Develop generalizable grasping/placing policies (RL/offline data, safe exploration) ▪ Sim-to-real transfer with domain adaptation; implement multiple gripper configurations; integrate tactile/force sensing where feasible ▪ Milestones: ▪ Policies demonstrated on multiple parts at the lab mock up ▪ Deliverables: ▪ Policy libraries and tactile integration components ○ WP 2.4: Dynamic racks: moving/swinging, safety, line-mock integration ▪ Months 25–30 ▪ Key Tasks: ▪ Real-time tracking of moving targets ▪ Visual servoing and compliant control during contact ▪ Safety interlocks and basic process monitoring ▪ Milestones: ▪ Dynamic rack tests show robust tracking at target speeds; safe engagement/recovery demonstrated in the lab mock up ▪ Deliverables: ▪ Dynamic validation report; updated interfaces and safety models ○ WP 2.5: Field pilot deployment and validation ▪ Months 31–36 ▪ Key Tasks: ▪ Field deployment planning; safety certification planning; operator training; maintenance plan ▪ Final validation on a real line; optimization based on field feedback ▪ Milestones: ▪ Field pilot in real environment completed ▪ Deliverables: ▪ Field pilot report; deployment package;

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Responsibilities, tasks and activities ○ Technical leadership: drive perception, planning, manipulation, and learning-based policies; define technical targets and success criteria. ○ Architecture and interfaces: design a modular software stack (perception, planning, manipulation) ○ Simulation and validation: develop digital twin, offline/online testing frameworks ○ Prototypes and algorithms: deliver perception modules, grasping/mounting strategies, motion-planning/control algorithms; provide reusable software libraries. ○ Integration and lab validation: implement test integration in a lab ○ Safety, risk, and compliance: perform risk analyses in cooperation with the Korean SME ○ Knowledge transfer: produce documentation, training materials, and conduct operator/engineer training; support handover to SME. • Resources ○ Dedicated project team ○ lab facilities with robots and 3D cameras ○ simulation software and infrastructure
Expected R&R of Korean SME Partner	• Responsibilities, tasks and activities ○ Provide the use case: Rack and part catalogue, operating conditions, safety requirements, and acceptance criteria. ○ Industrial pilots and evaluation: design and execute line mock-ups and real-line pilots; collect KPIs (throughput, uptime, quality, safety) and provide feedback. ○ Deployment support and system integration on site, especially regarding Korean regulations and processes ○ Potentially: System design and hardware development (e.g. for developing optimal versatile grippers ○ Test materials and line access: provide racks, bins, parts and access to production environments if needed; support on-site testing. ○ Operational validation: conduct field data collection, supervise pilot runs, and verify manufacturability and maintainability. ○ Risk and compliance support: supply real-world safety requirements, conduct risk assessments in cooperation with the Research Institute, and assist with certifications. ○ Deployment planning and training: help formulate rollout plans, operator training and maintenance procedures. • Resources

	○ Providing testing material (parts, racks, conveyors) ○ Engineering support ○ Safety/compliance personnel ○ IT infrastructure for data integration

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• **Technological achievements** ○ End-to-end robotic prototype for automated rack feeding and unloading in coating lines ○ Robust 3D perception for parts, hooks, and racks in cluttered, moving environments ○ Learning-based grasping and placing that generalizes across parts and randomly deformed hooks ○ Precise motion planning and control for hanging/unhanging under uncertainty, integrated in a modular system architecture ○ Validation in realistic painting/galvanizing environments with defined KPIs (throughput, success rate, uptime) • **Commercialization potential** ○ Basis for a marketable robotic cell for automatic rack handling in coating processes ○ Reusable software modules (perception, grasping, planning) for integration by robot OEMs and system integrators ○ Clear business case: addresses labor shortages, ergonomics, quality, and traceability demands in coating and related industries • **Industrial and economic impact** ○ Reduced manual, repetitive, and hazardous work; improved worker safety and ergonomics ○ Higher process stability, throughput, and coating quality through repeatable automation ○ Increased competitiveness and new business opportunities for partners in coating, surface treatment, and other rack-based applications .

6. References

(2) https://www.ipa.fraunhofer.de/en/reference_projects/Turnkey_solution_for_bin-picking_in_cooperation_with_Liebherr-Verzahntechnik_GmbH.html

(1) https://www.youtube.com/watch?v=8aYQsZpvIGU

(3) https://www.youtube.com/watch?v=eJ3iknylJ3o

(max 10 pages)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization